

The empirical recurrence for OEIS A299737: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A299737 records “Empirical recurrence of order 79”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 100$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 79, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 79 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A299737 is “Number of nX5 0..1 arrays with every element equal to 0, 1, 2, 3, 4, 5 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 512, 11303, 280642, 6896530, 170243005, 4203272237, 103785926879, ...

The entry states:

Empirical recurrence of order 79 (see link above)

As of the “Last modified” line on the live entry (Feb 18 2018 10:13:55, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 100$ of the 1024, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 100$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 200$ exact terms of the model returns order 79 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{79} c_i a(n-i),$$

c_1	28	c_{21}	−1389739513825	c_{41}	−414481026324764	c_{61}	48573935856010
c_2	−51	c_{22}	−2446778876311	c_{42}	1392268481605941	c_{62}	2899864164487
c_3	−677	c_{23}	3601069230117	c_{43}	1011007997263644	c_{63}	−15972736099280
c_4	−2914	c_{24}	12280942841627	c_{44}	−1203442420265826	c_{64}	−6021711328660
c_5	18755	c_{25}	−2524620479610	c_{45}	−1565194414062738	c_{65}	3737044273777
c_6	119014	c_{26}	−39784293254342	c_{46}	556266593944757	c_{66}	2203314715335
c_7	−287162	c_{27}	−25166909588311	c_{47}	1690104144221856	c_{67}	−419147736712
c_8	−2050233	c_{28}	82027997630013	c_{48}	169158761748338	c_{68}	−681930398571
c_9	141339	c_{29}	122412980690797	c_{49}	−1303430213150661	c_{69}	−15931106252
c_{10}	30953054	c_{30}	−80574126194876	c_{50}	−592177197067153	c_{70}	136715253064
c_{11}	31453281	c_{31}	−301039458673127	c_{51}	742020732143578	c_{71}	29818154552
c_{12}	−297369624	c_{32}	−62531668384158	c_{52}	639975049716776	c_{72}	−6021371488
c_{13}	−622007279	c_{33}	454102252634139	c_{53}	−286721133679790	c_{73}	−3605458976
c_{14}	1852085694	c_{34}	373473539731238	c_{54}	−494343451949987	c_{74}	−2399047744
c_{15}	7468317171	c_{35}	−437637442565695	c_{55}	15262634810618	c_{75}	−162804608
c_{16}	−7544393452	c_{36}	−720319008257575	c_{56}	305851366222045	c_{76}	463133184
c_{17}	−57165286794	c_{37}	248073124091948	c_{57}	84336424767196	c_{77}	−80553984
c_{18}	−4396874045	c_{38}	992740683460923	c_{58}	−127900327517915	c_{78}	−36126720
c_{19}	329710148744	c_{39}	15820388946137	c_{59}	−89160744769433	c_{79}	5455872
c_{20}	303657317766	c_{40}	−1226025703496765	c_{60}	30346345402908		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 79, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $79 + 4$ terms removed returns order 79 as well, so $\deg q_{\min} = 79 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $79 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 79 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A299737.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.